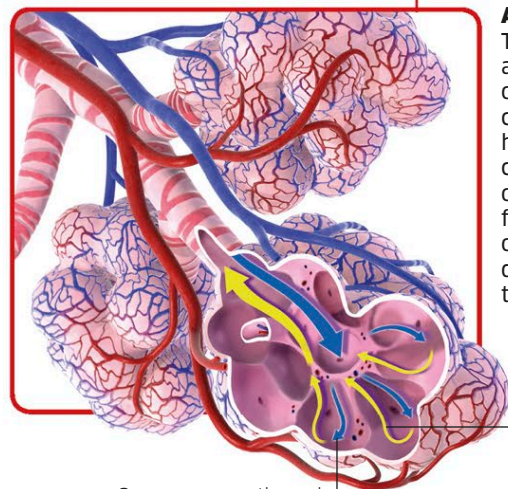
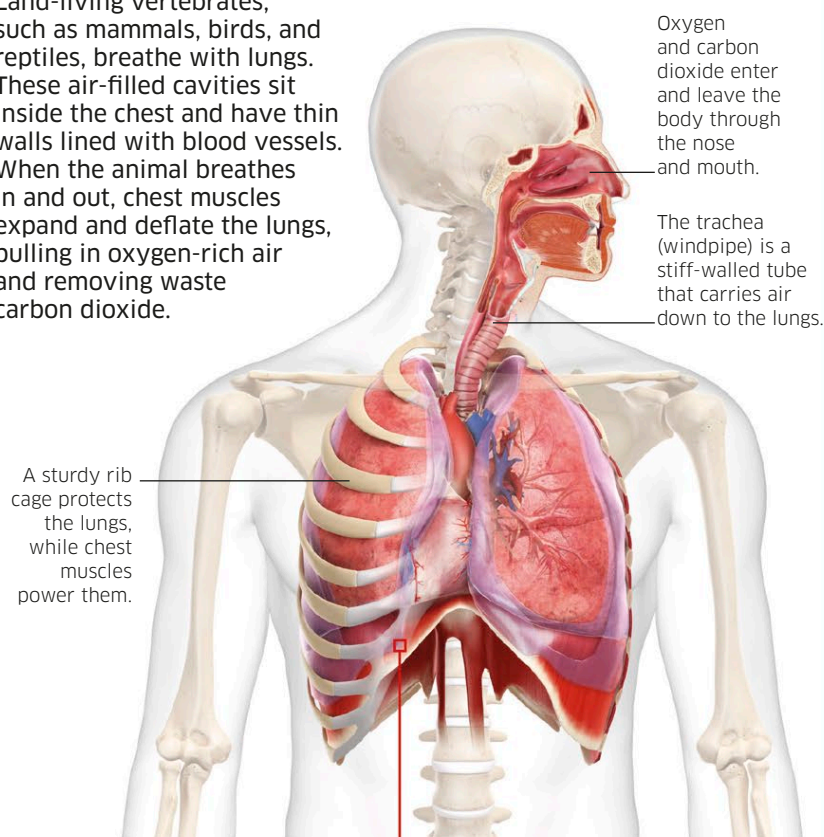


Breathing with lungs

Land-living vertebrates, such as mammals, birds, and reptiles, breathe with lungs. These air-filled cavities sit inside the chest and have thin walls lined with blood vessels. When the animal breathes in and out, chest muscles expand and deflate the lungs, pulling in oxygen-rich air and removing waste carbon dioxide.

**Alveoli**

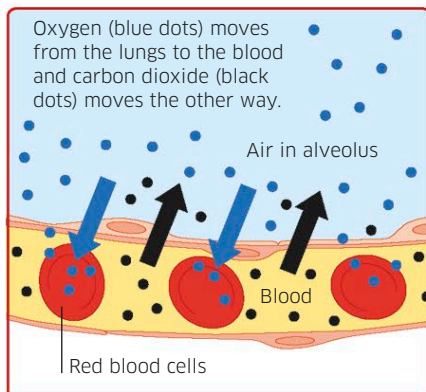
The lungs of mammals are made up of millions of microscopic sacs called alveoli. Each sac has an ultra-thin wall covered in a network of blood capillaries. This fine surface allows lots of oxygen and carbon dioxide to move between the air and the blood.

Oxygen moves through the very thin walls of the alveoli into the blood capillaries.

Waste carbon dioxide is transferred back from the blood to the lungs.

Diffusion

Oxygen and carbon dioxide are able to cross the microscopic membrane between the lungs and the blood by diffusion: a process by which molecules naturally move from an area where they are highly concentrated to one in which their numbers are fewer. This happens all around the body, as gases move between blood and respiring cells.



Breathing

An animal breathes to supply its cells with oxygen—a vital resource that helps to burn up food and release much-needed energy around the body.

All organisms—including animals, plants, and microbes—get energy from respiration, a chemical reaction that happens inside cells. Most do this by reacting food with oxygen, producing carbon dioxide as a waste product. To drive the oxygen into the body, different animals have highly adapted respiratory systems, such as lungs or gills. These can exchange large quantities of gas, carrying oxygen to respiring cells in the bloodstream, and excreting waste carbon dioxide.

Oxygen traveling in the blood is attached to a pigment called hemoglobin—the substance that gives blood its red color.

Breathing with gills

Gills are feathery extensions of the body that splay out in water so that aquatic animals can breathe. The delicate, blood-filled gills of fish are protected inside chambers on either side of their mouth cavity. A fish breathes by opening its mouth to draw oxygen-rich water over its gills. Some fishes rely on the stream created as they swim forward, but most use throat muscles to gulp water. Oxygen moves from the gills into the blood, while stale water emerges from the gill openings on either side of the head.

